

DETECTING DIRECTION VIOLATION ON SINGLE DIRECTION LANES BY VEHICLE USING HAAR CASCADE

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Digital Image Processing, CS

1. INTRODUCTION

Vehicle detection remains a challenging task in modern Intelligent Transportation System (ITS) and traffic control systems. Identification of vehicles and monitoring movement dynamically requires detecting visual appearance, determining position and tracking. Having these capabilities automatized would positively impact traffic safety without constant manual surveillance. As a result, real time image detection framework can be utilized as a tool to analyze vehicular traffic behavior and identify traffic violations.

Many main roads such in Jogja and in general are designed for single-directional traffic to maintain a smoother flow of traffic. However, many drivers still violate the traffic laws and go against the flow of traffic for the sake of their own benefits. This creates a massive safety issue for everyone on the road, often leading to traffic jams, if not, head-on collisions.

Identifying vehicles that move against the designated traffic direction can help improve road monitoring and support traffic safety. An automated detection system can reduce the need for continuous manual observation by detecting vehicle movement from CCTV footage. Therefore, this project aims to develop a simple system that can detect vehicles and identify whether they are moving in the correct or opposite direction of the traffic flow.

2. PROBLEM AND SOLUTION

2.1 Problem

Vehicles that violate traffic by going the opposite direction of the lane causes accidents that could result in severe injuries whilst disrupting the flow of traffic, as a single wrong-way vehicle could cause sudden braking, congestion, and secondary accidents (rear-ends, crashes, etc.). This problem is prevalent in the roads of Indonesia, in this case Yogyakarta, yet enforcement is still limited with the impossibility of traffic officers being present in every lane to watch for violations.

2.2 Solution

1. Haar Cascade detection: Scan each video frame with rectangular feature masks to detect vehicles from their typical contrast patterns (shadow under the car, headlights, windshield edges).
2. Center point tracking: Calculate the geometric center of each detected vehicle and follow its displacement across multiple frames.
3. Direction check: If a vehicle's displacement opposes the lane's default direction, mark it with a red box as a wrong-way vehicle.

3. METHOD

The program will be made using python with Google Colab, utilizing the opencv library for the image processing algorithms. The main algorithm we will use is the Haar Cascade object detection algorithm. We will obtain the dataset through the jogja cctv footage website: <https://cctv.jogjakota.go.id/>, kaggle, or other sources.

Haar Cascade itself is a machine learning based algorithm that scans the image using rectangular feature masks to detect contrast variations typical of vehicle structures, such as shadow under a car, headlights, or windshield edges.

To determine whether a vehicle is going against the one way current, we will use multiple frames of a video footage. When a vehicle is detected, the algorithm will calculate the geometric centerpoint of the vehicle. Then, it will track the displacement of that centerpoint across the multiple image frames, deciding whether it is against the current or not. For example, if in the footage, the default lane current would be to go up the frame (positive Δy), if a vehicle's displacement throughout the multiple frames is down (negative Δy), the program will mark it with a red box, indicating that this vehicle is going against the lane

current.

4. REFERENCE

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